

Pricing of a novel parametric weather derivative - A Multi-variate Spatial Basket

HKUST

1. Local Risk Exposure - Baseline Stochastic Models Fitted to local Data: $S = f(Y_l(t))$
 - ▶ Weather variables and Climate events - Class of Gauss-Markov processes suitable for indifference pricing measure.
 - ▶ Continuous-time AR(p) model, Seasonal volatility fit.
 - ▶ Generalized Hyperbolic Distribution fit.
 - ▶ Normal Inverse Gaussian fit.
2. Global Traded Index - Futures / Parametric Proxy: $Y_g(t)$
 - ▶ Constructing Real Estate investment portfolio and HDD, CDD, CAT futures index proxy under risk-neutral distribution.
 - ▶ Climate data for Compound Risk Indices.
3. Pricing + Hedging - Spatial Basket: $X(t)$ and $a(t)$ Strategy
 - ▶ Constructing synthetic future index.
 - ▶ Spatio-temporal climate model.

Biovariables and Morphology params

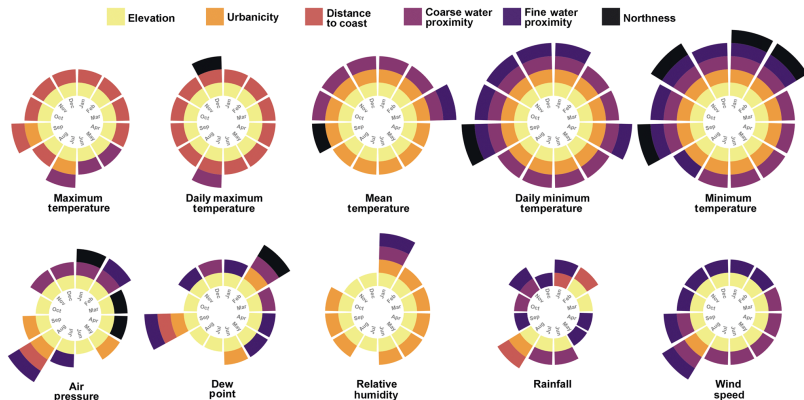


Figure: Reference variables

Real Estate Investment and weather index - Y_I ; Y_g

- Exposure model (since no market portfolio) Real Estate - Estate-wise aggregation: local climate exposure Y_I

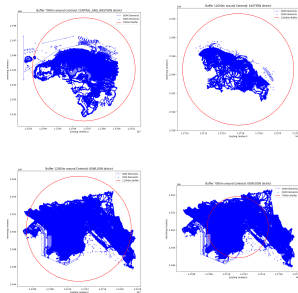


Figure: Intra-District aggregation

Futures on HDD, CDD, CAT, and PRIM are traded as indices.
Multivariate Index strategy based on using a linearization
based feedback law - synthetic global climate index Y_g

Pricing and Model Specification

$$F(t, y) \left(\alpha g(t, y) + \frac{1}{2} \frac{\mu^2(y)}{\sigma^2(y)} \right) = \frac{\partial F}{\partial t}(t, y) + GF \quad (1)$$

with terminal condition $F(T, y) = 1$ where we use proxies of y : Y_1 as the local risk and Y_2 as the global risk for payoff. Feynman-Kac-

$$F(t, y) = \mathbb{E}_y \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right].$$

$$V(t, x, y) = -e^{-\alpha x} \mathbb{E}_y \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right]$$

Trading strategy: $\phi_t = \frac{\mu(Y_1)}{(\sigma^2(Y_1)\alpha)}$

Temporal Dynamics

Temperature example - for basis risk as the source of market incompleteness

$$dT(t, x) = d\Lambda(t, x) - \alpha(x)(T(t, x) - \Lambda(t, x)) dt + \sigma(t, x) dW(t, x)$$

- ▶ $W(t, \cdot)$: $L^2(D)$ -valued Wiener process
- ▶ Continuous spatial covariance function $q(x, y)$, strictly positive definite and symmetric.
- ▶ Given weather index $I(x_i)$ at locations $x_1, \dots, x_n$.
- ▶ Optimal strategy $a(t)$ minimizing variance of error in replicating $I(y)$ at location y using spatial correlation and geometry of locations.
- ▶ Risk Neutral Time-Dynamics of Index Futures Prices;

$$dF_{\text{index}}(t, \tau_1, \tau_2) = \sigma(t) \int_{\tau_1}^{\tau_2} e_1' \exp(A(s - t)) e_p \Phi$$

Φ is the cumulative standard normal distribution

Fitting the Model

- Explicit solution of $X(t)$:

$$X(s) = \exp(A(s - t))x + \int_t^s \exp(A(s - u))e_p\sigma(u) dB(u)$$

- Temperature $T(t)$ is defined as:

$$T(t) = \Lambda(t) + X_1(t)$$

- $X_1(t)$: CAR(p) model, $\Lambda(t)$: seasonality function

1. Fit seasonal function $\Lambda(t)$ with least squares.
2. Fit AR(p) model on deseasonalized temperatures.
3. Fit seasonal volatility $\sigma(t)$ to residuals.

Contract Payoff and Pricing

Utility with a derivative written on the underlying process Y - akin to Asian structure and $K = 0$

$$f_1(Y) = \left(\int_0^T Y_t dt - K \right)^+; f_2(Y) = \left(\int_0^T \mathbb{1}_{\{Y_t > \epsilon\}} dt - K \right)^+,$$

$$f(Y) = \int_0^T h(Y_s) ds, \quad 0 \leq t' \leq t \leq T.$$

- ▶ Numerical solution of underlying optimization problem analysed through its dual under HJB (David)
- ▶ Explicit closed form solution to HJB: RN measure vs historical measure vs Indifference measure (Musiela)

Spatial Hedging - Weight Matrix - Global index Y_g

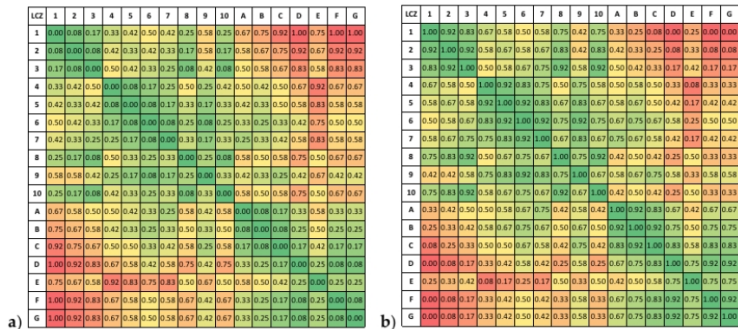


Figure 3. (a) Dissimilarity and (b) similarity metric for LCZ classes (Appendix A in [4]).

Figure: LCZ (Local Climate Zone) based wights matrix reference

Spatial Hedging - Weight Matrix - Global index Y_g

- ▶ Model with no spatial dependency in Λ , α , and σ .
- ▶ Spatial dependency modeled by a spherical correlation function (Benth 2012)

$$q(x, y) = 1 - \frac{3}{2} \frac{|x - y|}{\gamma} + \frac{1}{2} \frac{|x - y|^3}{\gamma^3}$$

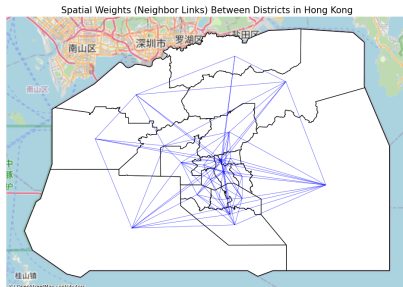


Figure: Rent Roll based